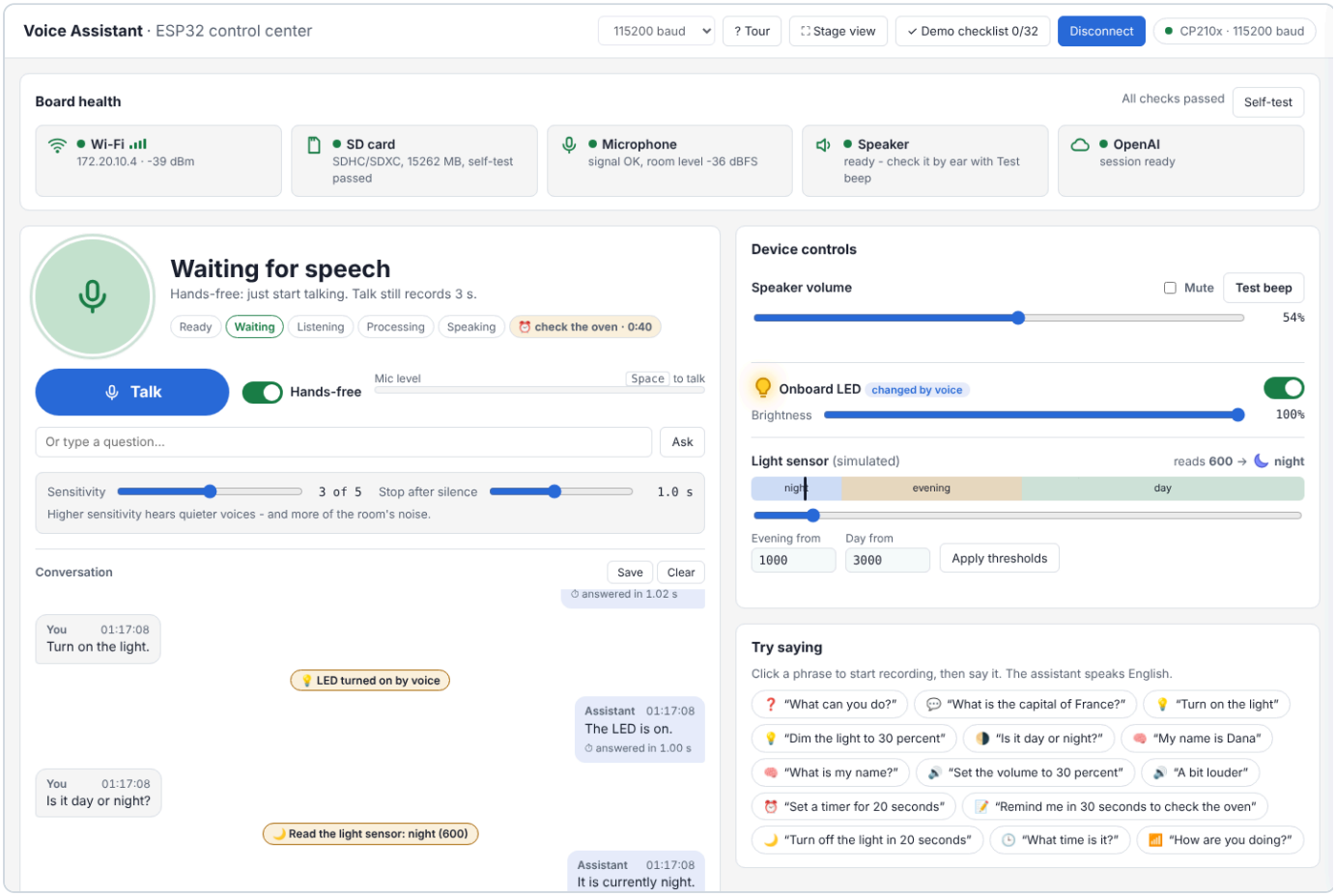
Quick Start Guide - Voice Assistant on ESP32

Group A2, IoT course, Summer 2026. One page for starting the system on your own. All the code and documents are in the GitHub repository (scan the QR code).

Getting started

1. The board is connected to the computer by USB, and the phone's hotspot is on (2.4GHz, and the board needs to know it - see "Connecting to the hotspot" below).
2. The control center (`control_center/control_center.html`) is open in Chrome. Click **Connect board** and select the board's COM port.
3. Wait until all the lights at the top are green. The assistant says an opening sentence. You can click **Self-test** for a quick check of all the parts.
4. Click **Talk** (or press the space bar) and speak, for up to 3 seconds. It works best in English; Hebrew works, but less well. Or turn on **Hands-free** and just speak.

New to the screen? The **Tour** button at the top gives a short tour.



Connecting to the hotspot

The assistant needs Wi-Fi, and in the demo we use a phone's hotspot, not the Technion network (the board did not work with it. We did not check why; our guess is WPA2-Enterprise login or a sign-in page). Do this once, before the demo.

1. On the phone, turn on a hotspot on **2.4GHz**. The board does not connect to 5GHz. On an iPhone: Personal Hotspot → **Maximize Compatibility**. On Android: Hotspot → Band → 2.4 GHz.

2. The network name is up to 32 characters, preferably in English, with no emoji and no apostrophe. The password is 8 to 64 characters.
3. The board learns the network in one of two ways:
   - **Without compiling:** in the control center, in the **Wi-Fi network** card (under Device controls), enter the name and password and click **Save & connect**. The card shows Sent and then Saved only after the board confirms, and the network stays saved on the board after a Reset. If you save while the board is still in its startup Wi-Fi wait, the card shows `Sent - the board reads it once its Wi-Fi wait ends (up to 20 s)…` and waits up to 40 seconds for Saved.
   - **In the code:** the name and password in `ESP32/SECRETS.h` (the default, as the README explains). **Back to default** in the card returns to it. If a network saved in the card fails twice in a row (for example a wrong password), the board switches to the network from `SECRETS.h` on its own.
4. Check it: Reset the board, and wait for the Wi-Fi light to turn green. Do this the day before, and again when you arrive, not at the last minute.
5. To switch networks (for example to demo "the network changed"): a second phone's hotspot, also on 2.4GHz, in the Wi-Fi network card.

What to try

Topic	What to say	What will happen
What it can do	`What can you do?`	It lists them in one sentence
General question	`What is the capital of France?`	It answers out loud within about a second
Light	`Turn on the light` then `Dim the light to 30 percent`	The LED turns on and dims, and the bulb on the screen glows to match
Volume	`Set the volume to 30 percent` then `A bit louder`	The slider moves and it confirms
Reminder	`Remind me in 30 seconds to check the oven`	A countdown, and at the end beeps, a blinking LED and `Time's up - check the oven`
Time	`What time is it?`	The time and date from the board's clock
Part of the day	`Is it day or night?`	According to the light slider (simulates a sensor), or according to **Auto day-cycle**: a full simulated day in about 2 minutes
Memory	`My name is Dana` then `What is my name?`	It remembers the name

What you see on the screen

- **The big circle:** what the assistant is doing right now. Green is ready, red is listening, orange is thinking, blue is speaking.
- **Conversation:** what it heard and what it answered, how long it took, and what it did on the board.
- **Device controls:** volume, mute, LED and the light sensor. The settings stay saved on the board after a Reset.
- **Statistics:** how many commands, response times and a graph. **Summary** shows everything in full screen.

- **Stage view** (the F key): the assistant shown large, for people standing far away.
- **On the board itself:** the BOOT button works as the Talk button, and there are status sounds, so you can use the board without the computer. Before a manual recording you hear a short high start tone, and a rejected press (for example in Offline mode) sounds as a low tone. Holding BOOT for 2 seconds is a technician check: 4 beeps in the order Wi-Fi, OpenAI, SD card, microphone (high = OK, low = failed).

What each color and message means: System states and error messages.

If something does not work

What you see	What to do
Talk shows Busy	The assistant is still speaking (also during the greeting, a timer message or an answer to a typed question), and Talk is rejected until it finishes. Wait for the end of the answer, or click **Cancel**
A red light on Wi-Fi or OpenAI, the circle shows Offline	Check the hotspot (and that it has internet). The board reconnects on its own, without a Reset, within seconds of the network coming back. A different network: Device controls → Wi-Fi network
A red SD light	There is no card, or it was removed while the board was running: the assistant works normally, just without recordings. Insert a FAT32 card and press Reset (EN). Putting the card back without a Reset leaves the light red
A red banner about USB	Connect the cable again. The board is powered by the cable, so it restarts (about 10 seconds until the greeting), and the control center reconnects on its own
Talk shows Downloading…	The board is sending a recording to the computer and is not listening in the meantime: about 3 seconds for an answer, up to about 40 seconds for a 10-second voice command. Talk and Ask come back when the download finishes
The status is gray with `Board not responding - no report for 15 s. If it stays like this, press Reset (EN) on the board.`	The board reports every 5 seconds, and 15 seconds without a report grays out the status. Wait a moment, and if it stays like this: Reset (EN) on the board
It did not understand	Speak closer, preferably in English, and check in the bubble what it heard. In a noisy room you can type the question in the field under Talk
Hands-free is set off by noise	Lower the Sensitivity, or use Talk